Power BI Dashboard Portfolio

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Project 1: Global UFO Sightings Dashboard

1.1 Visualisation 1

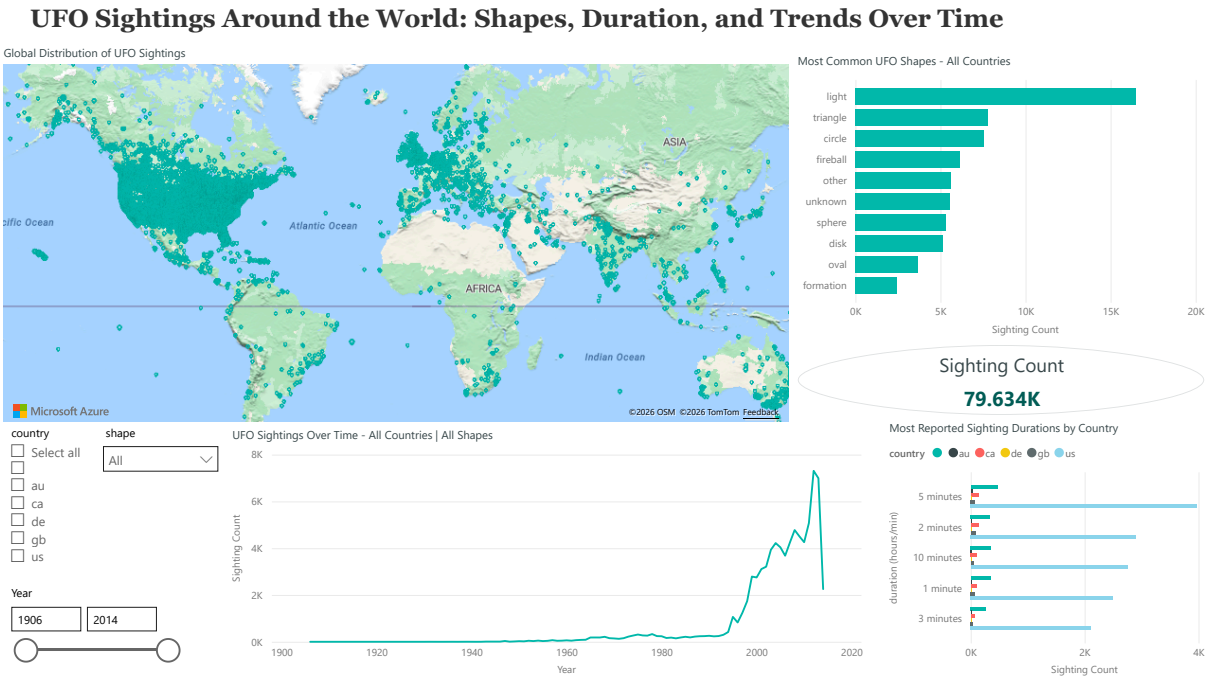

Figure 1: UFO Sightings Around the World: Shapes, Duration, and Trends Over Time

Link to the visualisation: Public PowerBi link

1.2 Report 1

The interactive dashboard on Figure 1 demonstrates UFO sightings around the world and shows the observed shapes of UFOs, the reported duration of sightings, and the years in which UFOs were reported more frequently. It also gives users the opportunity to filter the data by country, shape, and year range.

The main purpose of the dashboard is to explore how UFO sightings differ by location, shape, duration, and time. The main metric used in the dashboard is the total number of sightings. This is useful because it helps show whether sightings are more concentrated in some places, whether some shapes are reported more often than others, and how the number of reports changes over the years.

The charts and map can help answer specific questions such as: Where did the most sightings happen in the world? What are the most commonly observed UFO shapes? How long were sightings usually observed? In which years did reports become more frequent? The dashboard can be useful for enthusiasts interested in the topic, travellers who may be curious about frequently reported locations, and analysts who want to explore patterns in UFO data, such as where sightings are most commonly reported.

The dashboard consists of a map, where users can hover and see more detailed information reported, including sighting count, shape, city, duration, and comment. Because the dataset contains latitude and longitude data, the map was used to show the exact locations of sightings and to support geographic exploration. This makes the location aspect of the dataset more meaningful and easier to understand.

The dashboard also includes a bar chart showing the most common UFO shapes by total sighting count, such as light, triangle, and circle; a total sighting count card; slicer filters with checkboxes to select country; a year range filter from 1906 to 2014; a dropdown filter for shape; a time series chart showing UFO sightings over time; and a chart showing the most commonly reported sighting durations by country, such as 5 minutes, 2 minutes, and 10 minutes.

The layout was designed to be clear and intuitive, with the map shown first because it is the most suitable visual for this dataset and helps users immediately understand that location is one of the main parts of the analysis. The filters were placed nearby so users can easily see that they can explore the dashboard further by country, shape, and year range. This creates a logical reading flow and makes the dashboard understandable without needing the report. Appropriate chart types were chosen for each question: the map for geographic distribution, the bar chart for comparing shapes, and the line chart for trends over time. Colour contrast was kept clear and consistent so the visuals remain readable. Titles were written in plain language, and the dashboard title explains both the topic and what users can explore. Tooltips provide extra detail without overcrowding the visuals, and the layout avoids clutter by using spacing and clear separation between elements. These decisions also support usability and accessibility by making navigation easier, keeping text readable, and ensuring that information can be understood quickly.

Project 2: Global Fossil CO2 Emissions Dashboard

2.1 Visualisation 2

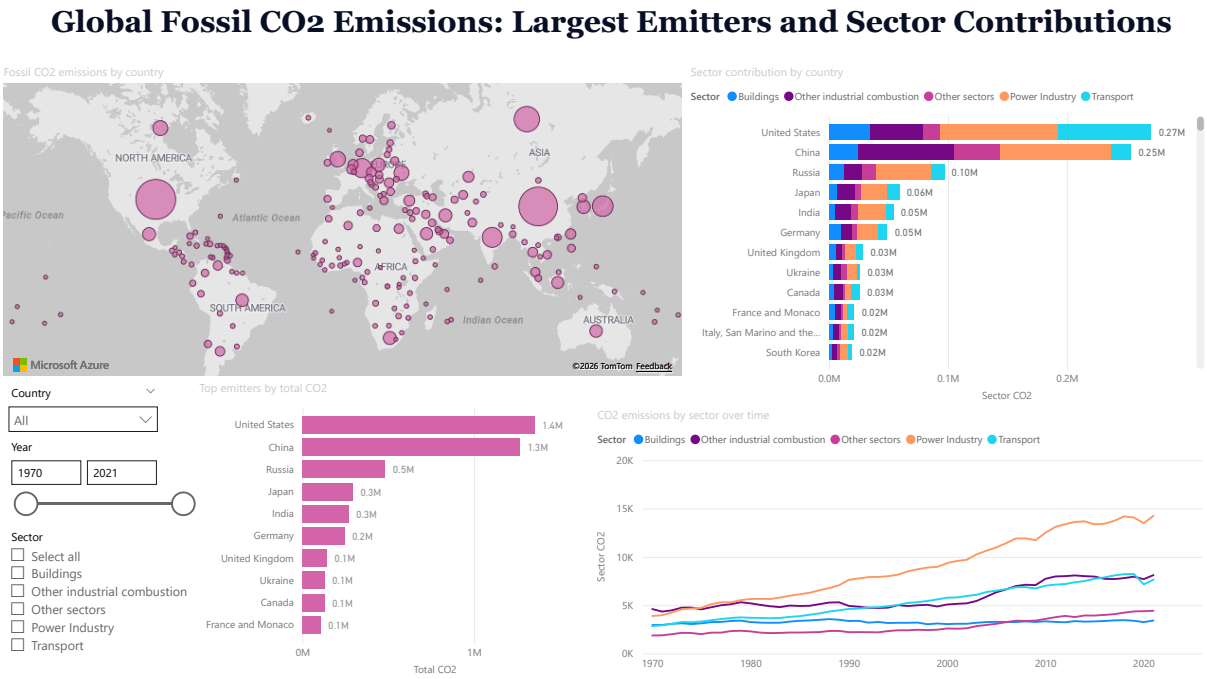

Figure 2: Global Fossil CO2 Emissions: Largest Emitters and Sector Contributions

Link to the visualisation: Public PowerBi link

2.2 Report 2

This interactive dashboard on Figure 2 was designed to explore global fossil CO2 emissions across countries and sectors. Its purpose is to help users identify which countries produce the largest fossil CO2 emissions, which sectors contribute most to these emissions, and how sector-related emissions have changed over time. The dashboard helps answer questions such as: Which countries are the largest contributors to fossil CO2 emissions? Which sectors contribute most to emissions in these countries? Which sectors have shown the greatest increase in emissions over the years? These insights can support students and analysts studying ecology, sustainability, and climate-related issues, as well as users interested in understanding global emission patterns.

A map was used to provide an immediate global overview of fossil CO2 emissions by country, because it is well suited for showing spatial distribution and helping users recognise geographical patterns. Bubble size represents the total level of emissions, which makes it easier to spot major emitters quickly and compare broad differences across the world. A simple map style was chosen to reduce visual distraction and keep the focus on the data rather than on decorative background elements. To complement the map, a bar chart of the top emitters was included because bar charts are more effective than maps for precise ranking and direct comparison between countries. This chart makes it easier to see that countries such as the United States, China, Russia, and Japan are among the largest contributors in the selected view. The layout was designed to support a clear reading order, beginning with the global overview map, followed by country ranking, and then more detailed sector-based comparisons and trends, so that the dashboard moves from broad context to more specific analysis.

A time series chart was used to show CO2 emissions by sector from 1970 to 2021. This makes it possible to observe long-term changes and identify which sectors have grown most over time. In the selected data, the power industry shows the strongest increase compared with the other sectors. A stacked bar chart was

also included to compare sector contribution by country. This visual helps users understand which sectors dominate emissions in different countries. For example, in the United States, power industry and transport make large contributions, while in China, power industry and other industrial combustion are especially important.

Interactive filters were added to support exploration. A country dropdown allows users to focus on one country, a sector checklist supports multiple selections, and a year range slider makes it possible to examine changes across time. Together, these interactions make the dashboard more flexible and meaningful.

Data Cleaning Reports

3.1 Data Cleaning Report 1

The dataset used for the dashboard was a scrubbed version of the original UFO sightings dataset, so only minimal transformation was required. A Year field was derived from the datetime column to support time-based analysis, and a Sighting Count measure was created for aggregation across visuals.

Several data quality issues were identified in the UFO sightings dataset before building the dashboard. First, some datetime values could not be parsed correctly because of invalid time formats such as 24:00. These values were adjusted to allow the column to be converted into a usable date/time format. Some missing values were found in fields such as country and state. Blank values were mostly left as they were, because in some rows the city field already contained country information while the country field was empty. These values were not manually filled in to avoid incorrect assumptions. A DAX measure was also added in Power BI to create dynamic titles, allowing some chart titles to change depending on the selected filters.

3.2 Data Cleaning Reports 2

Each sheet in the dataset was reviewed separately because the workbook contains multiple tables with different measures and structures. Year columns were unpivoted from wide to long format to improve filtering and interactivity in Power BI. Column names were standardised. Aggregate non-country entries, such as global totals and regional groups, were excluded from country-based visuals to keep comparisons clear and fair. Values were not heavily rounded, because stronger rounding would have distorted the relative differences between countries and sectors and reduced the accuracy of comparison.